

§11.9 Representations of Functions as Power Series

Homework : 1,7,9,11,13,15,29,37,38

- 源頭函數： $\frac{1}{1-x} = 1 + x + x^2 + \dots = \sum_{n=0}^{\infty} x^n$, $|x| < 1$.

(i) 直系函數：

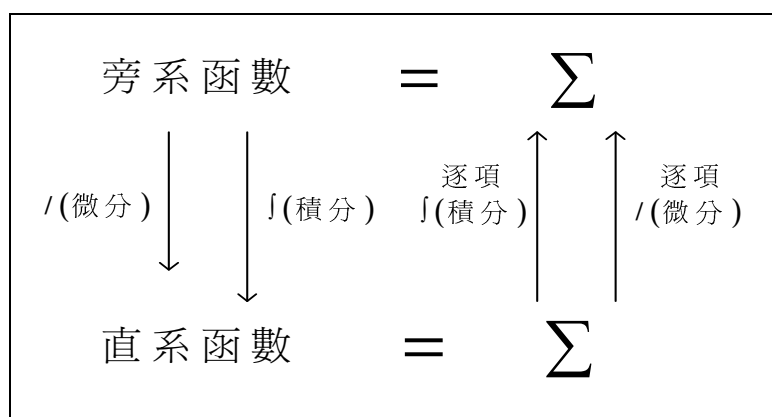
Example 1 : $\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, $|x^2| < 1 \Leftrightarrow |x| < 1$.

Example 2 : $\frac{1}{2-x} = \frac{1}{2} \frac{1}{1-\frac{x}{2}} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}}, |x| < 2.$

Example 3 : $\frac{x^3}{x+2} = \frac{x^3}{2} \frac{1}{1 - \left(-\frac{x}{2}\right)} = \frac{x^3}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+3}}{2^{n+1}}, |x| < 2.$

(ii) 旁系函數：(related to 直系函數 by 微分和積分)

圖解：



註：

1. Power series 在絕對收斂的範圍(即收斂區間)可作逐項微分或積分的動作。
2. 有時需微分或積分不止一次才到直系。

Example 4 : Find the power series representation of $\ln(1-x)$.

Solution :

$$\ln(1-x) = c - x - \frac{x^2}{2} - \frac{x^3}{3} \dots = c - \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$$

$$\begin{array}{ccc} \downarrow / & & \uparrow \int \\ \frac{-1}{1-x} & = - \sum_{n=0}^{\infty} x^n = -1 - x - x^2 \dots & |x| < 1 \end{array}$$

$$\text{令 } x=0 \Rightarrow \ln 1 = 0 = c. \text{ So,}$$

$$\ln(1-x) = - \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}, \quad |x| < 1.$$

$$= - \sum_{n=1}^{\infty} \frac{x^n}{n}, \quad |x| < 1.$$

Example 5 : $f(x) = \tan^{-1} x$

Solution :

$$\tan^{-1} x = c + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \quad |x| < 1.$$

$$\begin{array}{ccc} \downarrow / & & \uparrow \int \\ \frac{1}{1+x^2} & = \sum_{n=0}^{\infty} (-1)^n x^{2n} & |x| < 1 \end{array}$$

$$\text{令 } x=0 \Rightarrow \tan^{-1} 0 = 0 = c + 0 \Rightarrow c = 0.$$

$$\Rightarrow \tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| < 1.$$

Example 6 : $f(x) = \frac{1}{(1-x)^2}$

Solution :

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}, \quad |x| < 1.$$

$$\begin{array}{ccc} \downarrow \int & & \uparrow / \\ \frac{1}{1-x} & = \sum_{n=0}^{\infty} x^n, & |x| < 1. \end{array}$$

Example 7 : Evaluate $\int \frac{1}{1+x^7} dx$ as a power series.

Solution :

$$\begin{aligned} & \int \frac{1}{1+x^7} dx \\ &= \int \sum_{n=0}^{\infty} (-1)^n x^{7n} dx \\ &= \sum_{n=0}^{\infty} \int (-1)^n x^{7n} dx \\ &= \left(\sum_{n=0}^{\infty} (-1)^n \frac{x^{7n+1}}{7n+1} \right) + c \quad |x| < 1. \end{aligned}$$

Example 8 : (a) $\sum_{n=1}^{\infty} nx^{n-1} = ?$, $|x| < 1$.

(b) (i) $\sum_{n=1}^{\infty} nx^n = ?$, $|x| < 1$.

(ii) $\sum_{n=1}^{\infty} \frac{n}{2^n}$.

(c) (i) $\sum_{n=2}^{\infty} n(n-1)x^n = ?$, $|x| < 1$.

(ii) $\sum_{n=2}^{\infty} \frac{n^2 - n}{2^n} = ?$

(iii) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$.

Solution :

(8-a) For $|x| < 1$, we have

$$\sum_{n=1}^{\infty} nx^{n-1} = \sum_{n=1}^{\infty} (x^n)' = \left(\sum_{n=0}^{\infty} x^n \right)' = \left(\frac{1}{1-x} \right)' = \frac{1}{(1-x)^2}.$$

(8-b-i) For $|x| < 1$, we have

$$\sum_{n=1}^{\infty} nx^n = x \sum_{n=1}^{\infty} nx^{n-1} = \frac{x}{(1-x)^2} \quad (\text{此例子為旁系+直系}).$$

$$(8-b-ii) \quad \sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{\frac{1}{2}}{\left(1 - \frac{1}{2}\right)^2} = 2 \quad .$$

$$(8-c-i) \quad \text{For } |x| < 1, \text{ we have } \sum_{n=2}^{\infty} n(n-1)x^n$$

$$x^2 \sum_{n=2}^{\infty} n(n-1)x^{n-2} = x^2 \sum_{n=2}^{\infty} (nx^{n-1})' = x^2 \left(\sum_{n=1}^{\infty} nx^{n-1} \right)' = \frac{2x^2}{(1-x)^3}$$

$$(8-c-ii) \quad \sum_{n=2}^{\infty} \frac{n^2 - n}{2^n} = \frac{2\left(\frac{1}{2}\right)^2}{\left(\frac{1}{2}\right)^3} = 4 \quad .$$

$$(8-c-iii) \quad \sum_{n=1}^{\infty} \frac{n^2}{2^n} = \sum_{n=1}^{\infty} \frac{n^2 - n}{2^n} + \sum_{n=1}^{\infty} \frac{n}{2^n} = \sum_{n=2}^{\infty} \frac{n^2 - n}{2^n} + \sum_{n=1}^{\infty} \frac{n}{2^n} = 4 + 2 = 6$$